

Dylan L. Jow, PhD

Kavli Fellow | Kavli Institute for Particle Astrophysics and Cosmology
Stanford University
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Education

- 2018 - 2024 **Canadian Institute for Theoretical Astrophysics, University of Toronto**
PhD in Physics
Advisor: Dr. Ue-Li Pen
Thesis Title: "Theory and Applications of Wave Optics in the Gravitational and Plasma Lensing of Coherent Sources."
- 2014 - 2018 **University of British Columbia**
BSc. in Combined Honours Physics and Mathematics with Distinction

Academic Employment

- 2024 - present **Kavli Institute for Particle Astrophysics and Cosmology, Stanford**
Kavli Fellow

Collaborations and Affiliations

- 2024 - present **Canadian Hydrogen Intensity Mapping Experiment, Fast Radio Bursts (CHIME-FRB)**
- 2024 - present **Australian Square Kilometre Array Pathfinder Variables and Slow Transients (ASKAP-VAST)**
Principal Investigator of Extreme Scattering Event Monitoring Study
- 2025 - present **The Square Kilometre Array Observatory (SKAO) Science Working Group on the Transient Universe**

Teaching

- 2022 **University of Toronto, Department of Physics**
Instructor for PHY483, Relativity Theory I
- 2018 - 2024 **University of Toronto, Department of Physics**
Teaching Assistant
- 2015 - 2017 **University of British Columbia, Department of Mathematics**
Teaching Assistant

Selected Student Evaluations

"[Dylan] has provided some of the best teaching I've received throughout my four years at UofT and is very supportive and understanding of the class and its needs." – From PHY483 Student Evaluations, Fall 2022

"[Dylan] is an exceptional example of going to great lengths to ensure you learn the material, and then going a step beyond to do a little more." – From PHY483 Student Evaluations, Fall 2022

Professional Service

Ongoing	Journal Referee <i>MNRAS</i> <i>A&A</i> <i>ApJL</i> <i>PRL</i>
2026	SLAC Users Group HEP Funding Advocacy Trip to Washington, DC
2025 - present	KIPAC Seminar Committee
2025	Science Organizing Committee, Scintillometry Workshop 2025
2025	Local Organizing Committee (Stanford), Structure and polarization in the interstellar medium: A Conference in Honor of Prof. John Dickey
2021 - 2024	Canadian Institute for Theoretical Astrophysics Visitors Committee
2023	GPT-4 for Astronomy Exploratory Committee, CITA <i>Participated in pilot study of new generative AI technology in astronomy research. Reported findings to general CITA community.</i>
2022	Local Organizing Committee, Scintillometry Workshop 2022
2019 - 2020	Executive of the Physics Graduate Student Association at UofT <i>Organized social events and advocated for student concerns within the Department of Physics.</i>

Outreach and Lay Publications

2024 - present	Volunteer Adult Literacy Tutor for San Mateo County Library
2026	ComSciCon: Stories of Ethics, Engagement, and Dialogue in Science Case study writer
2024	Summer Undergraduate Research Program Workshop (UofT) "The Art of Scientific Writing", July 2 2024
2023	News article Dylan L. Jow, "Opinion: Grades are failing students", <i>The Varisty</i> , https://thevarsity.ca/2023/02/04/opinion-im-a-u-of-t-instructor-and-i-believe-we-need-to-abolish-grades/ , February 4, 2023
2022	Outreach Talk, Cosmos From Your Couch Dylan L. Jow, "Wonder and Awe in Astronomy", <i>Cosmos From Your Couch</i> , https://www.dunlap.utoronto.ca/events/wonder-and-awe-in-astronomy/ , June 18, 2022

Awards

Kavli Fellowship	2024
Van Kranendonk Teaching Assistant Award	2023
Mitacs Globalink Research Award	2022
NSERC Canada Graduate Scholarship – Doctoral	2021
Ontario Graduate Scholarship	2019
UofT Faculty of Arts and Science Admission Award	2018
Thomas and Evelyn Hebb Memorial Scholarship	2017
Stanley M Grant Scholarship in Mathematics	2017
J Fred Mui Memorial Scholarship in Science	2015
Governor General’s Academic Bronze Medal	2014

Publications and Presentations

I. Publications

Lead Author (pre-print)

Dylan L. Jow, Calvin Leung (2026), FRB scattering statistics through the CGM are sensitive to morphology and intermittency, arXiv, <https://arxiv.org/pdf/2603.00336>

Dylan L. Jow, Delon Shen (2026), Towards precision astrometry of scattered images of compact radio sources: scintillometry theory and prospects, arXiv, <https://arxiv.org/abs/2601.21041>

Lead Author (peer-reviewed)

Dylan L. Jow, Anna Tsai, Ue-Li Pen (2026), How many pixels are there in a polarized pulsar timing array map?, PhysRevD, Volume 113, <https://doi.org/10.1103/zvb6-7ggb>

Dylan L. Jow, Ue-Li Pen (2025), Measuring Cosmic Expansion with Diffractive Gravitational Waves, PRL, Volume 134, Issue 13, <https://doi.org/10.1103/PhysRevLett.134.131001>

Jow, D. L., Wu, X., & Pen, U.-L. (2024). Refractive lensing of scintillating FRBs by subparsec cloudlets in the multiphase CGM. Proceedings of the National Academy of Science, 121(39), <https://doi.org/10.1073/pnas.2406783121>

Jow, D. L., Pen, U.-L., & Baker, D. (2024). On the cusp of cusps: a universal model for extreme scattering events in the ISM. MNRAS, 528(4), 6292–6301. <https://doi.org/10.1093/mnras/stae300>

Jow, D. L., Pen, U.-L., & Feldbrugge, J. (2023). Regimes in astrophysical lensing: refractive optics, diffractive optics, and the Fresnel scale. MNRAS, 525(2), 2107–2124. <https://doi.org/10.1093/mnras/stad2332>

Dylan L. Jow, Ue-Li Pen (2022), Measuring lens dimensionality in extreme scattering events through wave optics, MNRAS. Volume 514, Issue 3, Pages 4069-4077. <https://doi.org/10.1093/mnras/stac1652>

Dylan L. Jow, Fang Xi Lin, Emily Tyhurst, Ue-Li Pen (2021), Imaginary images and Stokes phenomena in the weak plasma lensing of coherent sources, MNRAS. Volume 507, Issue 4, November 2021, Pages 5390-5402. <https://doi.org/10.1093/mnras/stab2337>

Dylan L. Jow, Simon Foreman, Ue-Li Pen, Wei Zhu (2020), Wave effects in the microlensing of pulsars and FRBs by point masses, MNRAS. Volume 497, Issue 4, October 2020, Pages 4956-4969. <https://doi.org/10.1093/mnras/staa2230>

Dylan L. Jow, Douglas Scott (2020), Re-evaluating evidence for Hawking points in the CMB, JCAP, Volume 2020. <https://doi.org/10.1088/1475-7516/2020/03/02>

Dylan L. Jow, Dagoberto Contreras, Douglas Scott, Emory F. Bunn (2019), Taller in the saddle: constraining CMB physics using saddle points, JCAP. Volume 2019. <https://doi.org/10.1088/1475-7516/2019/03/031>

Dylan L. Jow, Ryley Hill, Douglas Scott, J.D. Soler, P.G. Martin, M.J. Devlin, L.M. Fissel, F. Poidevin (2018), An application of an optimal statistic for characterising relative orientations, MNRAS. Volume 474, Issue 1, Pages 1018-1027. <https://doi.org/10.1093/mnras/stx2736>

Co-authorship

Tsai, A.C., Jow, D.L., Pen, U.-L. (2025), Reaching diffraction-limited localization with coherent PTAs, arXiv, <https://arxiv.org/pdf/2512.10795>

Manisha, C., Fortunato, J.A.S., Hagstotz, S., James, C.W., Jahns-Schindler, J.N., Jow, D.L. ... Wang, Y. (2026) Probing the Baryon Distribution with Fast Radio Bursts, arXiv, <https://arxiv.org/pdf/2606.29388>

Manisha, C., Fortunato, J.A.S., Hagstotz, S., James, C.W., Jahns-Schindler, J.N., Jow, D.L. ... Wang, Y. (2026) Fast Radio Bursts as Cosmological Probes, arXiv, <https://arxiv.org/pdf/2606.27714>

Shin, Kaitlyn, Leung, Calvin, Simha, Sunil, Andersen, Bridget C., Fonseca, Emmanuel, Nimmo, Kenzie, Bhardwaj, Mohit, Brar, Charanjot, Chatterjee, Shami, Cook, Amanda M., Gaensler, B. M., Joseph, Ronniy C., Jow, Dylan, ..., Stairs, Ingrid (2025) Investigating the sightline of a highly scattered FRB through a filamentary structure in the local Universe. ApJ, 993(2), 208. <https://doi.org/10.3847/1538-4357/ae093b>

Job Feldbrugge, Dylan L. Jow, Ue-Li Pen (2025), Complex classical paths in quantum reflections and tunneling, PhysRevD, 111(8), <https://journals.aps.org/prd/abstract/10.1103/PhysRevD.111.085027>

Ng, C., Pandhi, A., Mckinven, R., Curtin, A. P., Shin, K., Fonseca, E., Gaensler, B.M., Jow, D.L. ... Smith, K. (2025). Polarization Properties of 28 Repeating Fast Radio Burst Sources with CHIME/FRB. ApJ, 982(2), 154. [10.3847/1538-4357/adb0bc](https://doi.org/10.3847/1538-4357/adb0bc)

Leung, C., Jow, D., Saha, P., Dai, L., Oguri, M., & Koopmans, L. V. E. (2025). Wave Optics, Interference, and Decoherence in Strong Gravitational Lensing. SS, 221(2), 29. <https://doi.org/10.1007/s11214-025-01157-7>

Kempski, P., Li, D., Fielding, D. B., Quataert, E., Phinney, E. S., Kunz, M. W., Jow, D.L., Philippov, A. A. (2024). A Unified Model of Cosmic Ray Propagation and Radio Extreme Scattering Events from Intermittent Interstellar Structures. ApJ, 990(1), L18. <https://doi.org/10.3847/2041-8213/adde55>

Tsai, A., Jow, D. L., Baker, D., & Pen, U.-L. (2024). Scintillated microlensing: Measuring cosmic distances with fast radio bursts. Phys Rev D, 110(4), 043503. <https://doi.org/10.1103/PhysRevD.110.043503>

Job Feldbrugge, Dylan L. Jow, Ue-Li Pen (2023), Crossing singularities in the saddle point approximation, arxiv, <https://arxiv.org/abs/2309.12427>

F.X. Lin, R.A. Main, Dylan L. Jow, D.Z. Li, U.L. Pen, M.H. Van Kerkwijk (2023), Plasma lensing near the eclipses of the Black Widow pulsar B1957+20, MNRAS. Volume 519, Issue 1, Pages 121-135. <https://doi.org/10.1093/mnras/stac3456>

II. Presentations

Invited Talks

“Probing the physics and morphology of the cool CGM with FRB scattering”
Seminar at UCSB, 2026

“Scintillation, plasma lensing, and extreme scattering of compact radio sources”
COGs Seminar, KIPAC, 2025

“Scintillometry: Leveraging Interstellar Scintillation of Compact Radio Sources for Cosmology”
Cosmology Seminar, Albert Einstein Institute, 2024

“Radio Scattering by Refractive Sheets in the Ionized ISM”
Scintillometry Workshop, Florida Space Institute, 2024

“From Dust to Dust: Lessons from Scintillation and Radio Scattering”
Making Light of the Universe, University of British Columbia, 2024

“Wave lensing for precision cosmology”
Coffee talk at the Department of Astrophysical Sciences, Princeton University, 2023

“Real-time tunnelling through complexified path integrals.”
String Theory Seminar at National Taiwan University, 2023

“Prospects for gravitational and plasma lensing of FRBs”
Lunch Talk at ASIAA, 2023

“Cusps of cusps: a universal model for extreme scattering in the ISM”
Seminar at the Theoretical Astrophysics Including Relativity and Cosmology Institute, Caltech, 2022

“Wave optics in astrophysical lensing: unlocking the potential of the coherent sky”
Seminar at the Mullard Space Science Laboratory, 2022

“Wave optics in astrophysical lensing: unlocking the potential of the coherent sky”
Seminar at the Berkeley Centre for Cosmological Physics, 2022

“Wave optics in astrophysical lensing: unlocking the potential of the coherent sky”
Brown Bag Lunch Talk at MIT, 2022

Contributed Talks

“Probing the physics and morphology of the cool CGM with FRB scattering”
FRB 2026, National Astronomical Observatories, CAS, 2026

“Physical origins of Extreme Scattering Events”
Dynamic Radio Sky, University of Sydney, 2025

“Diffractive Gravitational Scintillation of NanoHertz Gravitational Waves”
Lensing and Wave Optics in Strong Gravity Workshop, ESI, 2024

“Cusps of cusps: a universal model for extreme scattering in the ISM”
Fields, Flows, and Filaments in the Magnetic ISM Workshop, Stanford University, 2024

“Where have all the lenses gone? Scattering of gravitationally lensed FRBs.”
FRB Conference at National Chung Hsing University, 2023

“Regimes in astrophysical lensing: refractive optics, diffractive optics, and the Fresnel scale”
Scintillometry Workshop, University of Toronto, 2022

“Wave Optics in Gravitational Lensing”
Scintillometry Conference, Max Planck Institute for Radio Astronomy, 2019